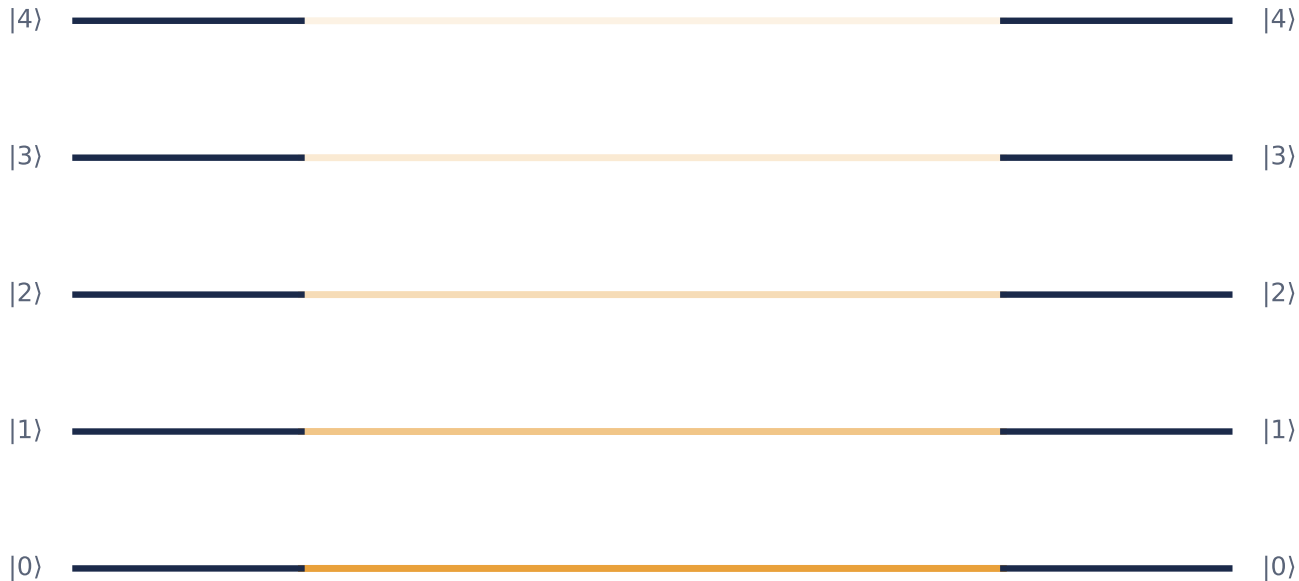


L

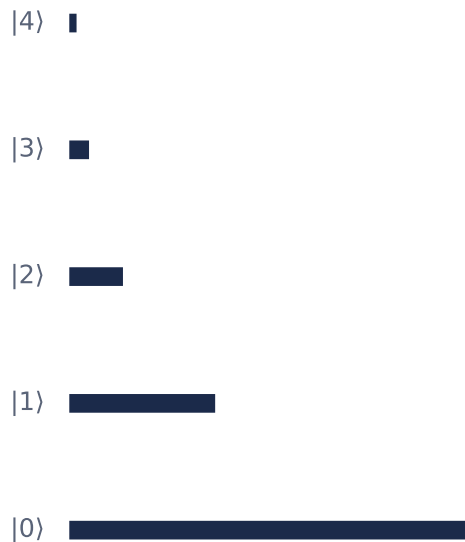
R



link opacity $\propto e^{-\beta E_n/2}$

$$|\text{TFD}(\beta)\rangle = Z^{-1/2} \sum_n e^{-\beta E_n/2} |n\rangle_L |n\rangle_R$$

Tr_R



$$\rho_L = e^{-\beta H}/Z$$

a thermal state at temperature $1/\beta$